

Where customer lifetime value comes from, why churn stays high, and what fixing delivery would unlock.

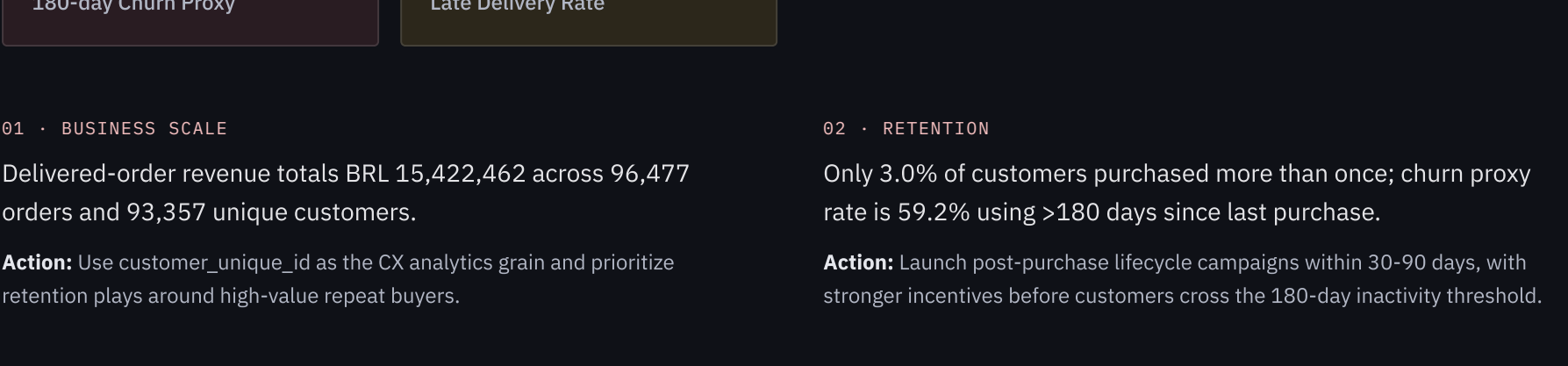
A statistical read of 96K delivered orders and 93K customers across 2016–2018.

CLV regression, churn classification, RFM segments, cohort retention, and a 12-month sales forecast — with the caveats spelled out.

CHAPTER 01 · EXECUTIVE READ

The headline numbers and what they ask of us.

Olist moved R\$ 15.4M across 96K orders, but only 3 in 100 customers came back. Late deliveries hit 8% of orders, and the churn proxy (180-day inactivity) sits at 59%. The opportunity isn't acquisition — it's keeping the customers we already paid to acquire.



01 · BUSINESS SCALE

Delivered-order revenue totals BRL 15,422,462 across 96,477 orders and 93,357 unique customers.

Action: Use customer_unique_id as the CX analytics grain and prioritize retention plays around high-value repeat buyers.

02 · RETENTION

Only 3.0% of customers purchased more than once; churn proxy rate is 59.2% using >180 days since last purchase.

Action: Launch post-purchase lifecycle campaigns within 30–90 days, with stronger incentives before customers cross the 180-day inactivity threshold.

03 · RFM SEGMENTATION

The largest revenue segment is Loyal high value with BRL 9,661,171; Champions average BRL 309 Monetary value.

Action: Protect Champions with premium service and early access; target Potential loyalists with cross-sell bundles and free-shipping thresholds.

04 · DELIVERY EXPERIENCE

Late deliveries occur on 8.1% of delivered orders; average review score is 4.16/5.

Action: Prioritize proactive delivery alerts, seller SLA monitoring, and recovery offers for late orders to reduce churn risk.

05 · MODEL PERFORMANCE

CLV regression $R^2=0.989$; order value regression $R^2=0.308$; churn model $AUC=0.636$.

Action: Use model scores for prioritization, not automated decisioning; validate on more recent holdout data before production use.

06 · FORECAST

Next 12-month forecast totals BRL 18,586,387, implying 56.2% vs the latest comparable 12-month sales baseline. Forecast method: Additive trend + month seasonality fallback (ModuleNotFoundError: Prophet unavailable).

Action: Use the forecast as a planning baseline and refresh monthly; install Prophet in the environment for the canonical Prophet model if the fallback was used.

CHAPTER 02 · MODELS AT A GLANCE

Three models, three different jobs.

Linear regression predicts CLV from order behavior. Logistic regression flags likely churners. Multiple regression predicts the value of a future order. Train/test split is 80/20.

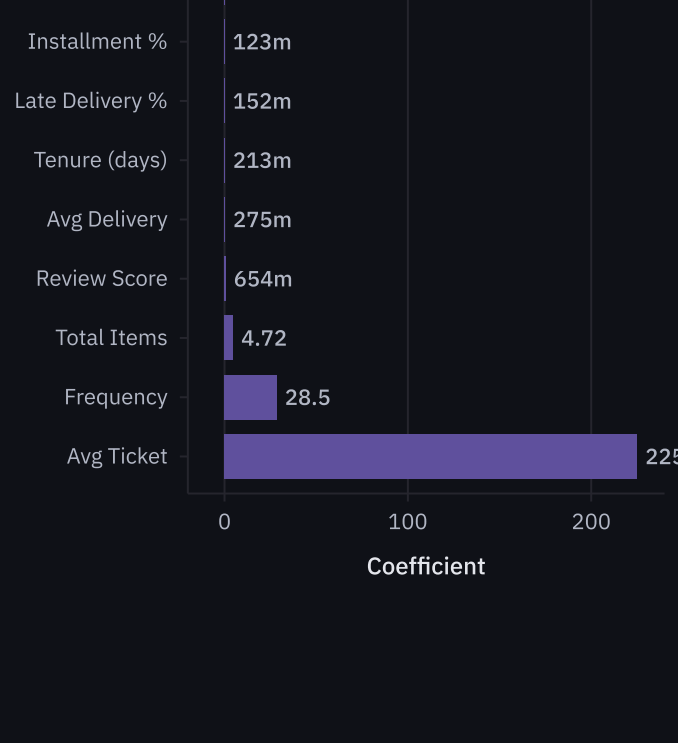
Model	Target	Rows	Train	Test	RMSE	MAE	R ²	Accuracy	F1	AUC
Linear regression - CLV	clv	93,357	70,017	23,340	22.57	5.22	0.99	--	--	--
Logistic regression - Churn	churned	93,357	70,017	23,340	--	--	--	0.62	0.73	0.64
Multiple regression - Order value	payment_value	96,477	72,357	24,120	176.92	83.39	0.31	--	--	--

CLV is nearly fully explained by order-level features ($R^2 \approx 0.99$), which is expected — CLV is largely arithmetic on past payments. Order-value prediction is harder ($R^2 \approx 0.31$), and the churn classifier separates the two classes modestly ($AUC \approx 0.64$). Treat the churn model as a triage tool, not a verdict.

CHAPTER 03 · CUSTOMER LIFETIME VALUE

What drives a customer's total spend.

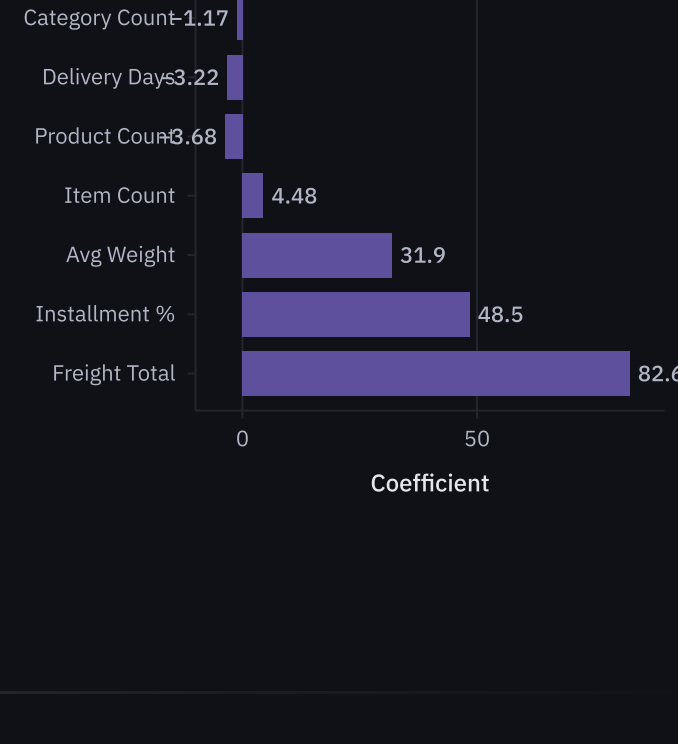
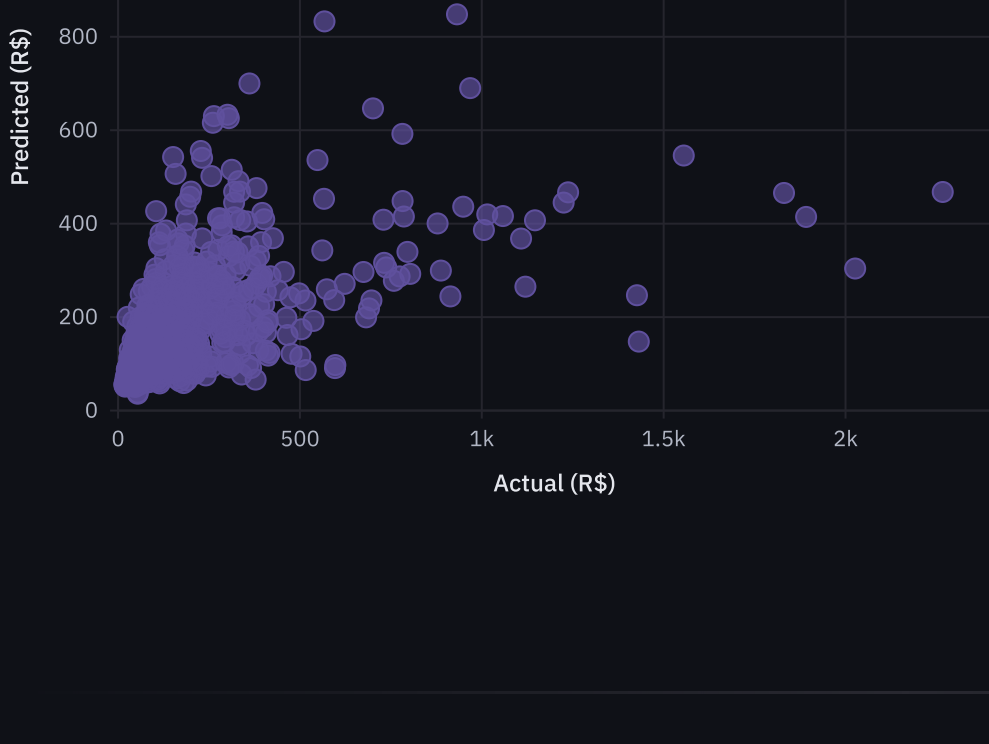
Linear regression on order Frequency, average ticket, tenure, payment behavior, and review signals.



CHAPTER 04 · ORDER VALUE

Pricing the next basket.

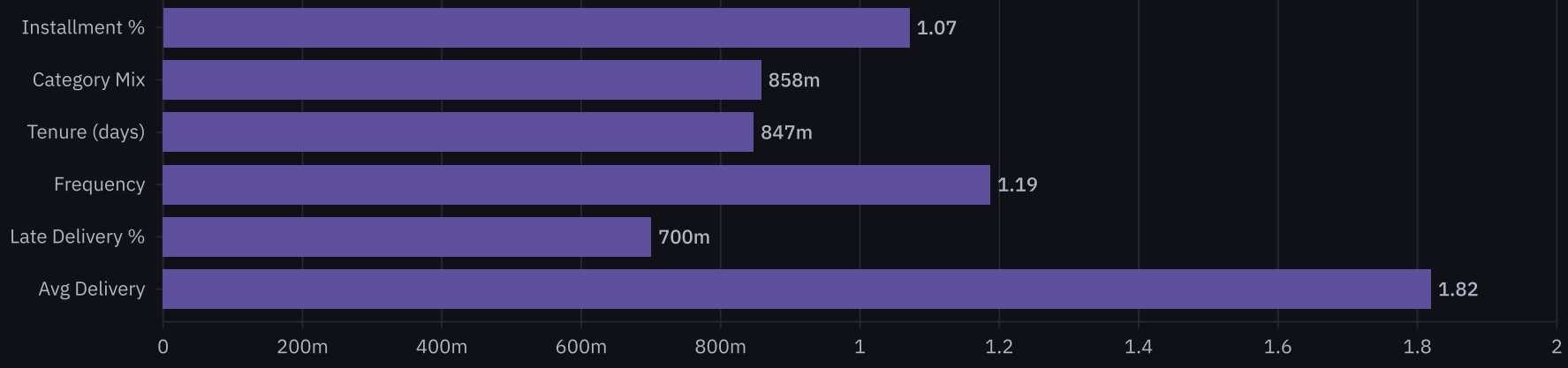
Multiple regression on freight, item count, category mix, payment installments, and customer geography.



CHAPTER 05 · CHURN RISK

Who is unlikely to come back.

Logistic regression on recency, satisfaction, delivery experience, and basket characteristics. Odds ratios > 1 push toward churn; < 1 protect against it.

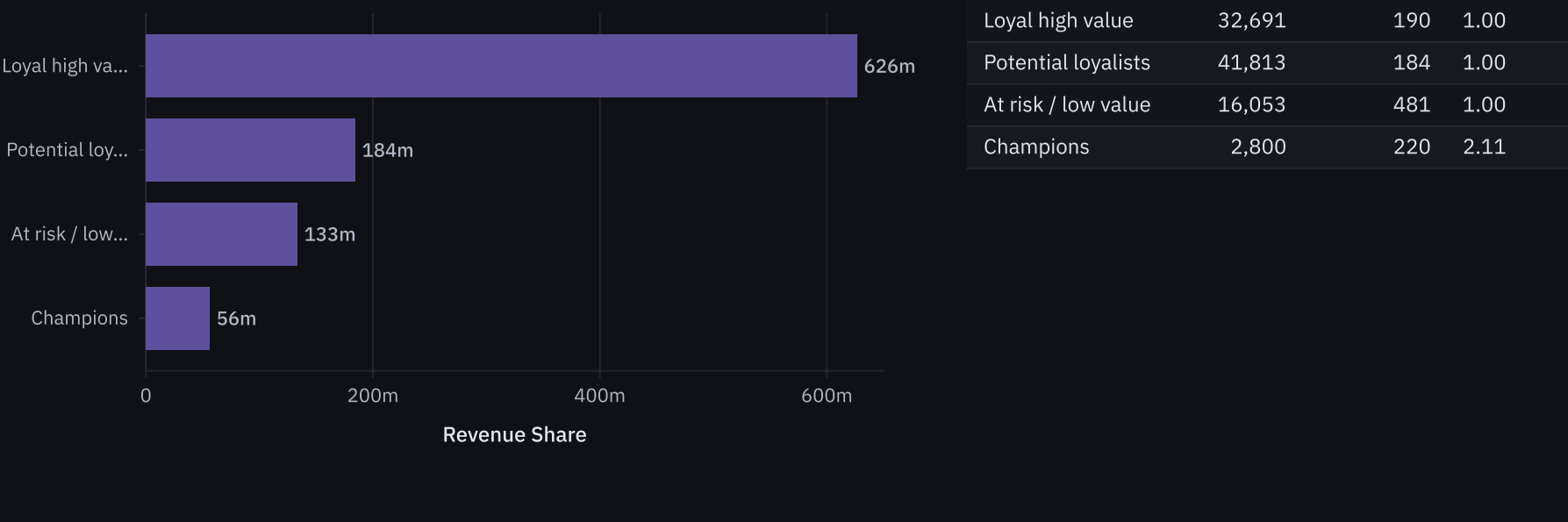


AUC of 0.64 is meaningful but not decisive — useful for ranking customers by risk so retention spend goes to the top decile, not for binary decisions.

CHAPTER 06 · RFM SEGMENTATION

Four customer worlds, very different economics.

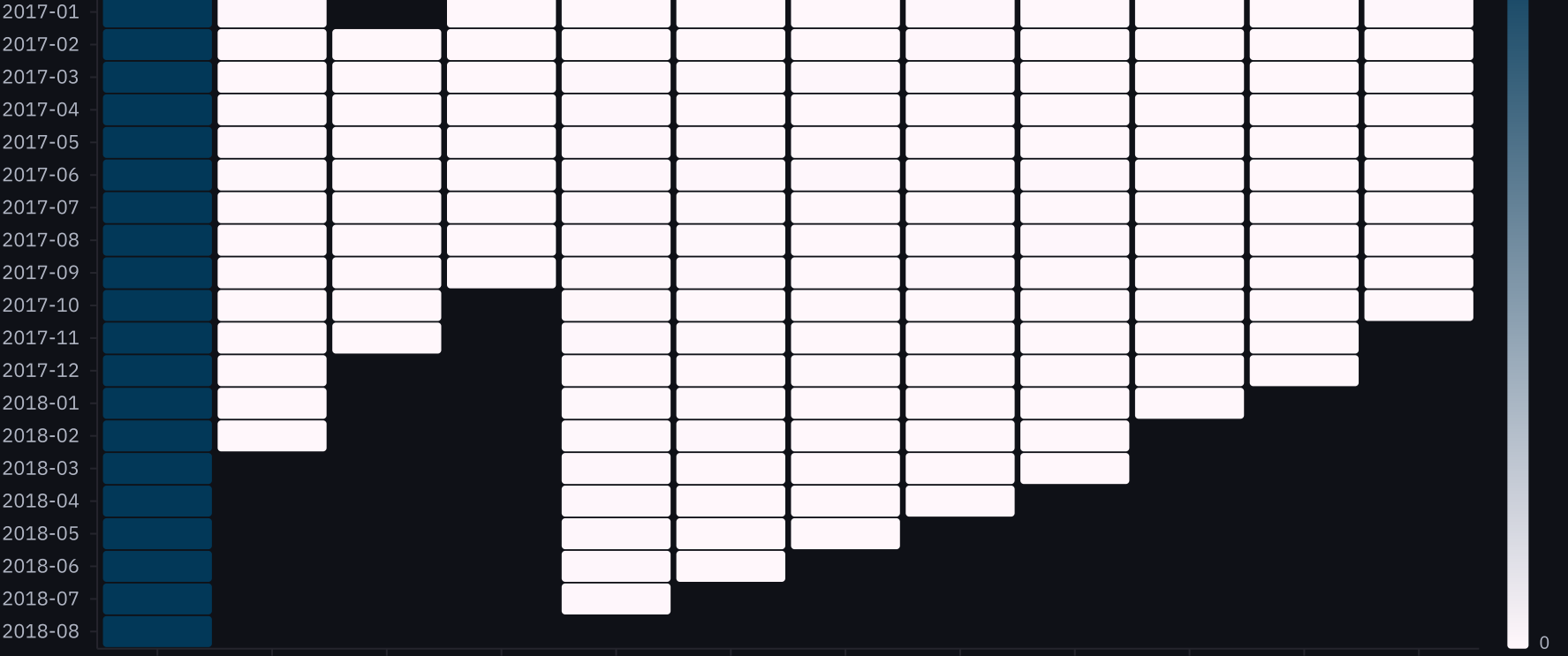
K-means clustering on recency, Frequency, and Monetary value. Each segment has a distinct retention and satisfaction profile.



CHAPTER 07 · COHORT RETENTION

Retention by acquisition month, first 12 months out.

Each row is a cohort of customers who placed their first order in that month. Each column is months elapsed since acquisition. Color intensity is the share of the cohort still active.



The cohort grid confirms the headline: month-2 retention is in the low single digits across every cohort. This is a one-and-done marketplace by default. Any program that lifts month-2 retention by even 1 point compounds significantly.

CHAPTER 08 · 12-MONTH SALES FORECAST

Where revenue trends if nothing changes.

Time-series projection on monthly sales totals through July 2019.

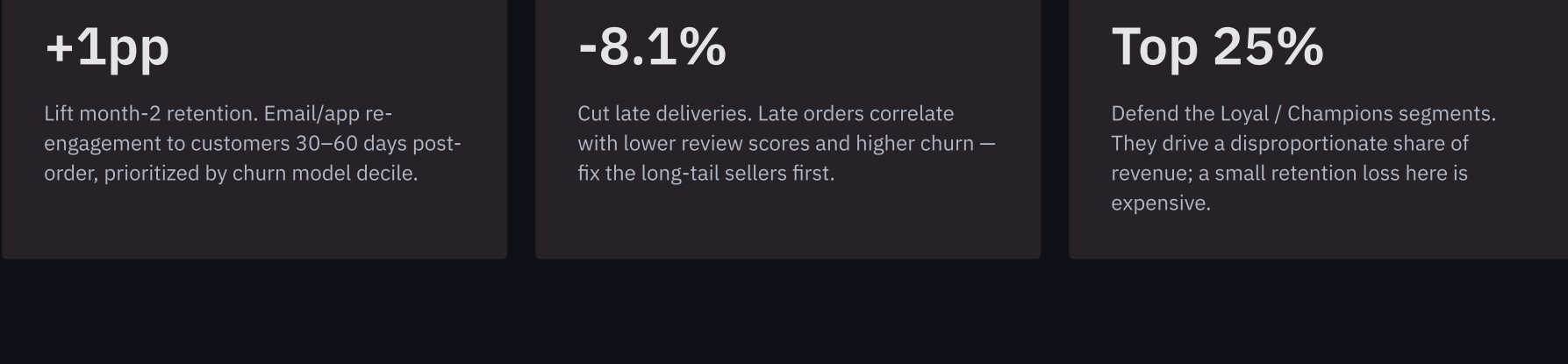


Forecast model: Additive trend + month seasonality fallback (ModuleNotFoundError: Prophet unavailable)

Methodology note. Prophet was not available in the analysis environment, so the forecast uses an additive trend plus monthly-seasonality fallback fit to the historical series. Treat the projection as a directional baseline, not a calibrated forecast — the confidence band is wider than what a fully tuned Prophet model would produce.

CHAPTER 09 · WHAT TO DO ABOUT IT

Three moves the data supports.



ARTIFACTS

Source files loaded: olist_geolocation_dataset.csv, olist_products_dataset.csv, olist_order_reviews_dataset.csv, olist_order_payments_dataset.csv, olist_sellers_dataset.csv, olist_order_items_dataset.csv, product_category_name_translation.csv, olist_customers_dataset.csv, olist_orders_dataset.csv

CSV artifacts written: executive_summary.csv, model_metrics.csv, model_coefficients.csv, rfm_segments.csv, segment_summary.csv, cohort_retention.csv, monthly_sales.csv, forecast.csv, plot_files.csv

PNG artifacts written: 01_clv_regression_actual_vs_predicted.png, 02_churn_logistic_calibration.png, 03_order_value_multiple_regression.png, 04_kmeans_rfm_segments.png, 05_cohort_retention_heatmap.png, 06_monthly_sales_forecast.png

Output directory: olist_customer_analytics_outputs

Forecast model used: Additive trend + month seasonality fallback (ModuleNotFoundError: Prophet unavailable)